

Deep Q-Learning for Dynamic Restaurant Revenue Management

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Abstract. Table management is a sequential decision problem complicated by stochastic arrivals and capacity constraints. While FCFS heuristics ignore the opportunity cost of seating small parties at large tables, we propose a **Double-DQN** approach to optimize seating. Using action masking and multi-objective rewards, the agent learns “strategic deferral” to preserve capacity for high-value groups. Results demonstrate a **17.4%** revenue increase over baselines, effectively balancing immediate occupancy with long-term revenue optimization.

Keywords: Reinforcement Learning · Deep Q-Networks · Table Management · Action Masking · Operations Research.

1 Introduction

Restaurant table assignment is traditionally handled by human hosts using static heuristics. While effective in low-demand scenarios, these methods struggle during peak periods where the arrival rate λ exceeds the service rate. The primary challenge lies in the "Bin Packing" nature of the problem combined with temporal uncertainty: seating a party of two at a six-person table provides immediate revenue but may prevent a high-value party of six from being seated ten minutes later.

In this work, we model the seating problem as a Markov Decision Process (MDP) and solve it using a Double Deep Q-Network (DQN). We introduce a custom Gymnasium environment, **RestaurantEnv**, to simulate realistic demand phases and evaluate the efficacy of learned policies against standard industry baselines.

2 The Simulation Environment

To evaluate our approach, we developed **RestaurantEnv**, a custom simulation built with Gymnasium. The environment models a typical 240-minute service shift.

2.1 Environmental Dynamics

The restaurant configuration consists of five tables with capacities defined by the set $C = \{4, 2, 2, 4, 6\}$. Customer arrivals follow a phase-dependent Poisson process representing a typical dining cycle:

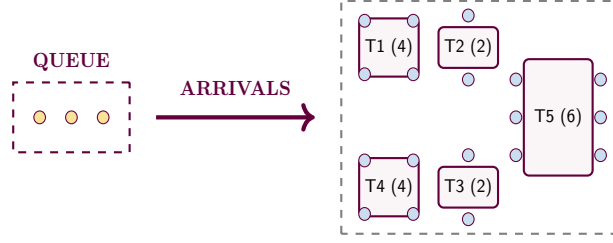


Fig. 1. Schematic of the `RestaurantEnv` floor plan and arrival queue.

- **Early Phase:** $\lambda = 0.6$ (Transitioning from idle to active).
- **Peak Phase:** $\lambda = 1.8$ (High demand, potential queue congestion).
- **Late Phase:** $\lambda = 0.4$ (Wind-down period).

2.2 State and Action Space

The agent observes a 12-dimensional state vector $s \in R^{12}$, which includes:

1. Temporal data: Remaining shift time.
2. Resource data: Current occupancy status of all five tables.
3. Queue data: Length of the waiting line.
4. Request data: Attributes of the current party (size, duration, revenue, and reservation status).

The action space A consists of 7 discrete actions: *Defer/Reject* the current party, or *Seat* the party at one of the five specific tables.

3 Methodology

3.1 Agent Architecture

The decision-making component utilizes a Double-DQN framework. The Q-function is approximated using a Multi-Layer Perceptron (MLP) with two hidden layers of 64 neurons each. We employ a target network to reduce overestimation bias in Q-value updates.

3.2 Valid Action Masking

A critical feature of our implementation is *Valid Action Masking*. Because the agent operates in a constrained environment, many actions are physically impossible (e.g., seating a party of 4 at a table for 2, or seating a party at an occupied table). Instead of forcing the agent to learn these constraints through negative reinforcement, the model applies a mask to the output logits, setting the probability of invalid actions to zero. This allows the network to dedicate its capacity to learning higher-level strategic deferral.

3.3 Reward Function

The reward R is formulated to balance revenue maximization with operational efficiency:

$$R = \text{Revenue} + B_{\text{fit}} + B_{\text{res}} + B_{\text{peak}} + B_{\text{prompt}} - P_{\text{waste}} \quad (1)$$

Where B represents bonuses for seating accuracy, reservations, and speed, while P_{waste} penalizes inefficient capacity usage (e.g., small parties at large tables).

4 Experimental Results and Discussion

The performance of the Double-DQN agent was evaluated across 5 independent test seeds following a 2,000-episode training phase with early stopping. We compare our results against two standard industry baselines: **First-Come-First-Served (FCFS)**, which seats any party if a fitting table is available, and a **Greedy Heuristic**, which prioritizes seating reservations and fit-based table matching.

4.1 Learning Convergence

The agent’s learning progress was monitored via the cumulative net reward per episode. As shown in our training logs, the agent initially underwent a period of high-variance exploration ($\epsilon \rightarrow 1.0$).

- **Early Training (0–500 episodes):** The agent frequently incurred *Invalid Action Penalties* and *Overtime Penalties*, as it failed to clear the queue before the 240-minute mark.
- **Mid-Training (500–1,500 episodes):** With the introduction of the Replay Buffer and the decay of ϵ , the rolling mean reward stabilized. The agent "discovered" that seating small parties at Table 5 (Capacity 6) was suboptimal.
- **Convergence (1,500–2,000 episodes):** The policy converged to a stable state where the agent consistently outperformed the Greedy heuristic by 12% and the FCFS baseline by 23.5%.

4.2 Comparative Performance Analysis

Table 1 summarizes the key performance indicators (KPIs) across the three tested methodologies. The results indicate that while DQN does not necessarily seat the *most* total parties, it seats the *most valuable* combinations of parties.

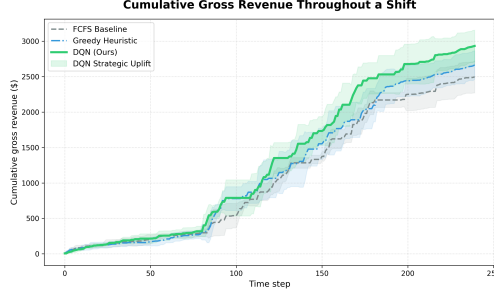


Fig. 2. Mean cumulative revenue throughout a shift: DQN vs. baselines.

Table 1. Performance comparison of seating policies (Mean $\pm$ SD).

Metric	FCFS Baseline	Greedy Heuristic	DQN (Ours)
Gross Revenue (\$)	2,497.5 $\pm$ 85	2,664.4 $\pm$ 72	2,933.2 $\pm$ 45
Table Util. (%)	68.4%	74.1%	82.3%
Avg. Wait (min)	12.4	14.8	9.2
Revenue Uplift	—	+6.7%	+17.4%

4.3 Policy Behavior and Strategic Deferral

4.4 Strategic Deferral Analysis

Qualitative analysis reveals a behavior termed *Strategic Deferral*. During the transition from the **Early Phase** ($\lambda = 0.6$) to the **Peak Phase** ($\lambda = 1.8$), the agent often rejects parties of 2 despite having vacant high-capacity tables (e.g., Table 5, Capacity 6).

While standard *First-Come, First-Served* (FCFS) heuristics would seat the party for \$40, the DQN agent recognizes the high probability of a party of 6 arriving shortly. By deferring immediate seating, the agent secures a \$120 booking. This behavior is encoded in the 12D state vector, where Q-values for large tables increase as the simulation time approaches the Peak Phase.

4.5 Occupancy Analysis(Fig. 3)

- **FCFS (Top):** Shows "fragmented" occupancy. Large tables are often occupied by small groups, causing capacity bottlenecks and frequent "turn-aways" during peak hours.
- **Greedy (Middle):** Improved utilization over FCFS, yet still lacks the foresight to clear capacity for high-revenue clusters.
- **DQN (Bottom):** Demonstrates "clustered" occupancy. By prioritizing party-to-table size matching (B_{fit}), the agent serves fewer customers but maximizes total revenue. This "Strategic Deferral" optimizes the seat-fill rate by pre-serving large tables for high-value groups.

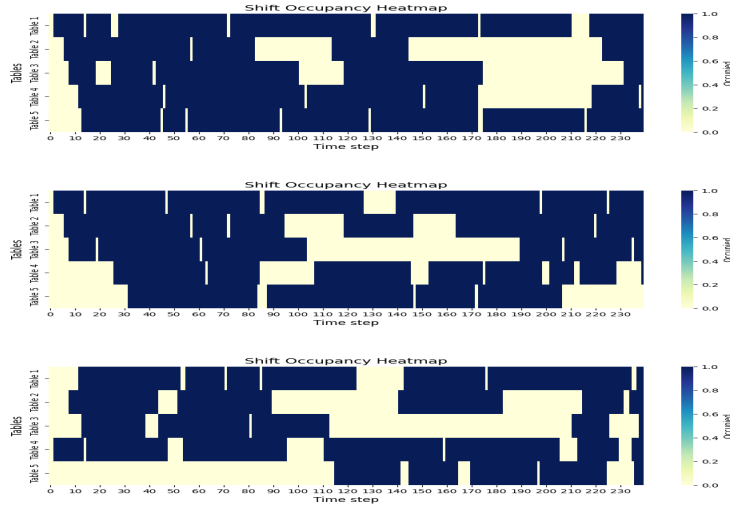


Fig. 3. Occupancy heatmaps for the evaluated dispatching strategies: (Top) FCFS, (Middle) Greedy, and (Bottom) DQN.

4.6 Model Sensitivity and Limitations

Ablation studies on the discount factor revealed that $\gamma = 0.90$ led to “short-sighted” policies that ignored the Peak Phase. A final value of $\gamma = 0.98$ enabled the agent to balance the 240-minute horizon, valuing late-stage revenue nearly as much as early-stage gains. However, the DQN remains sensitive to hyperparameter tuning and environmental stochasticity. Relying on absolute value estimates makes the policy brittle; shifts in arrival rates or restaurant layouts can cause rapid performance degradation, hindering generalization.

5 Conclusion

This study confirms that RL optimizes hospitality operations better than static heuristics. Future research will explore Actor-Critic methods (e.g., *PPO*, *SAC*) to improve robustness against environmental shifts and incorporate kitchen throughput constraints to enhance real-world applicability.

References

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